Forecasting the Variance of the Closing Half Hour for 0DTE SPX Options

Bullet-point draft of the full paper

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1 Related work

- **Realized variance and the HAR model:** $\mathbb{E}_t[\mathrm{RV}_{t+1}^{(d)}] = \beta_0 + \beta_d \mathrm{RV}_t^{(d)} + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)}$, with $\mathbb{E}_t$ the expectation given the information at the forecast origin on day t (here 15:30 ET) and $\mathrm{RV}_t^{(d)}$, $\mathrm{RV}_t^{(w)}$ and $\mathrm{RV}_t^{(m)}$ the average daily realized variance over the last 1, 5 and 22 trading days. Our HAR components $\bar{y}_t^{(k)}$ take the same averages over k intraday intervals of y, the square root of realized variance divided by its diurnal adjustment (Section 2).
  Corsi (2009), Andersen et al. (2003), Barndorff-Nielsen and Shephard (2004), Andersen et al. (2007), Corsi et al. (2008), Bollerslev et al. (2016), Bekierman and Manner (2018), Bennedsen et al. (2022), Hansen et al. (2012), Andersen et al. (2006).
  - Panel HAR, estimated jointly across assets, the cross-asset counterpart of our pooled intraday model, which pools the intraday intervals of one index: Bollerslev et al. (2018).
  - Barndorff-Nielsen and Shephard (2002) (realized variance).
- **Diurnal pattern:** volatility follows a U-shaped intraday pattern; the standard adjustment divides by a diurnal adjustment, here $y_{t,i} = \sqrt{\mathrm{RV}_{t,i}/s_{t,i}^2}$, with $\mathrm{RV}_{t,i}$ the realized variance of intraday interval i on day t and $s_{t,i}^2$ its diurnal adjustment.
  Andersen and Bollerslev (1997) (intraday periodicity), Engle and Gallo (2006) (intraday measures as indicators of daily volatility), Hizmeri and Bevilacqua (2025) (morning volatility uncertainty).
  - Taylor and Xu (1997) (diurnal adjustment), Engle and Sokalska (2012) (intraday volatility).
- **0DTE options and the variance risk premium:** the variance risk premium, implied minus expected realized variance, $\mathbb{E}_t^{\mathbb{Q}}[\mathrm{RV}_t] - \mathbb{E}_t^{\mathbb{P}}[\mathrm{RV}_t]$ under the risk-neutral ($\mathbb{Q}$) and physical ($\mathbb{P}$) measures, with RV_t the realized variance to expiration, is positive on average.
  Hizmeri and Bevilacqua (2025); option pricing and implied volatility Black and Scholes (1973), Merton (1973).
  - 0DTE options: Bandi et al. (2023), Beckmeyer et al. (2023), Brogaard et al. (2023), Dim et al. (2024), Vilkov (2026), Almeida et al. (2025); the variance risk premium: Carr and Wu (2009), Bollerslev et al. (2009).
- **Machine learning for volatility and returns:** a nonlinear function $f(\boldsymbol{x}_t)$ of a vector $\boldsymbol{x}_t$ of many predictors.

Gu et al. (2020) (returns), Christensen et al. (2023) (realized volatility); gradient-boosted trees Ke et al. (2017) (LightGBM), Chen and Guestrin (2016) (XGBoost); hyperparameter optimization Akiba et al. (2019) (Optuna).

  - Breiman (2001) (random forests), Friedman (2001) (gradient boosting); realized volatility: Bucci (2020), Rahimikia and Poon (2020), Zhang et al. (2024).

- **Forecast evaluation:** QLIKE is robust to noise in the volatility proxy: with a conditionally unbiased proxy it ranks forecasts as the true conditional variance would, and $\arg\min_{v>0} \mathbb{E}_t[\text{QLIKE}(\text{RV}_t, v)] = \mathbb{E}_t[\text{RV}_t]$, with $\text{QLIKE}(\text{RV}, v) = \text{RV}/v - \log(\text{RV}/v) - 1$.
  Patton (2011) (robust loss functions), Diebold and Mariano (1995) with the small-sample correction of Harvey et al. (1997), Hansen et al. (2011) (model confidence set), Ehm et al. (2016) (Murphy diagrams, elementary scores), Mincer and Zarnowitz (1969) (forecast-efficiency regressions).

  - Hansen (2005) (superior predictive ability), White (2000) (reality check), Romano and Wolf (2005) (StepM), Newey and West (1987) (HAC variance).

- **Shrinkage with many weak signals:**

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda(\alpha\|\boldsymbol{\beta}\|_1 + \tfrac{1-\alpha}{2}\|\boldsymbol{\beta}\|_2^2)$$

  for a response vector $\boldsymbol{y}$, a predictor matrix $\boldsymbol{X}$ and a penalty strength λ (ridge $\alpha = 0$, lasso $\alpha = 1$); multi-penalty ridge gives each block of predictors its own penalty, $\lambda \sum_j \omega_j \beta_j^2$ with the penalty weight ω_j constant within a block.
  Giannone et al. (2021) (the illusion of sparsity), Shen and Xiu (2025) (weak signals), Kozak et al. (2020) (shrinking the cross-section), Hoerl and Kennard (1970) (ridge), Tibshirani (1996) (lasso), Zou and Hastie (2005) (elastic net), van de Wiel et al. (2021) (multi-penalty ridge), van Wieringen (2023).

  - Zou (2006) (adaptive lasso, penalty weights), Friedman et al. (2010) (elastic-net parametrization).

- **Variable importance:** Shapley additive explanations (SHAP) decompose each forecast additively, $\hat{y}_t = \phi_0 + \sum_{j=1}^{p} \phi_{t,j}$, with $\phi_{t,j}$ the contribution of predictor j of p and ϕ_0 the average forecast.
  Lundberg and Lee (2017) (SHAP), Lundberg et al. (2018) (TreeSHAP), Aas et al. (2021) (dependent features).

2 Data

- **Sources**.
  - **E-mini S&P 500 futures (ES),** nearly 24 hours a day, 1998–2024, sampled on a 30-minute grid. Trading day $t = 1, \ldots, T$; intraday interval $i = 1, \ldots, M$, the half-hour intervals of the day; the closing interval $i = c$ is the closing half hour (15:30–16:00 ET).
  - $r_{t,i,k}$: the k-th one-minute log return in interval i of day t, $k = 1, \ldots, N_{t,i}$, with $N_{t,i} \leq 30$. Realized variance (Andersen et al., 2003):

$$\text{RV}_{t,i} = \sum_{k=1}^{N_{t,i}} r_{t,i,k}^2.$$

– Other realized measures, same interval:

$$\sum_k r_{t,i,k}, \quad \sum_k |r_{t,i,k}|, \quad \sum_k r_{t,i,k}^3, \quad \sum_k r_{t,i,k}^4, \quad \mathrm{RS}_{t,i}^+ = \sum_k r_{t,i,k}^2 \, \mathbf{1}\{r_{t,i,k} > 0\},$$

$$\mathrm{BV}_{t,i} = \sum_{k \geq 2} |r_{t,i,k}| \, |r_{t,i,k-1}|, \qquad \mathrm{AC}_{t,i} = \sum_{k \geq 2} r_{t,i,k} \, r_{t,i,k-1},$$

plus trading volume and the observation count $N_{t,i}$. The third and fourth sums are the realized skewness and kurtosis of Amaya et al. (2015) without their normalizations (division by $\mathrm{RV}_{t,i}^{3/2}/\sqrt{N_{t,i}}$ and by $\mathrm{RV}_{t,i}^2/N_{t,i}$). $\mathrm{RS}_{t,i}^+$ is the positive realized semivariance (Barndorff-Nielsen et al., 2010); $\mathrm{BV}_{t,i}$ is the bipower variation of Barndorff-Nielsen and Shephard (2004) without its $\pi/2$ scaling; $\mathrm{AC}_{t,i}$ is the first-order realized autocovariance.

– **Cross-section of index constituents,** equal- and value-weighted (EW, VW), 1993–2024: six return moments, turnover, order-flow imbalance (signed turnover) and bid-ask spreads. Order-flow imbalance, for each weighting ($\mathrm{OFI}^{\mathrm{EW}}$, $\mathrm{OFI}^{\mathrm{VW}}$):

$$\mathrm{OFI}_{t,i} = \frac{\mathrm{Buy}_{t,i} - \mathrm{Sell}_{t,i}}{\mathrm{Buy}_{t,i} + \mathrm{Sell}_{t,i}} \in [-1, 1], \qquad \text{missing if } \mathrm{Buy}_{t,i} + \mathrm{Sell}_{t,i} = 0,$$

with $\mathrm{Buy}_{t,i}$ and $\mathrm{Sell}_{t,i}$ the buy and sell turnover in interval i of day t.

– **StockTwits** attention and sentiment measures, from 2009 and 2010.

– **Cboe** volatility indices: VIX (from 2006), VIX3M (from 2009) and VVIX (from 2012), all to February 2024.

– Cboe option-volume series (the volatility-demand group), 2011 to August 2023.

– **FOMC** announcement times, 1993 to November 2023.

– **SPXW option quotes on expiration days** (S&P 500 Weeklys), every 30 minutes: bid, ask, midquote, underlying level, implied volatility and delta. 1,291 expirations, 2020–2025.

– *[name the vendor of each file].*

- **Intraday grid and cleaning.**

– Grid: 30-minute intervals from the first observed realized variance. Interval i of day t is retained if and only if its realized variance is observed and it lies outside the weekend break:

$$(t,i) \in \mathcal{G} \iff \mathrm{RV}_{t,i} \text{ observed and } (t,i) \notin (\text{Fri } 20{:}00,\ \text{Sun } 18{:}30),$$

with $\mathcal{G}$ the interval grid, the set of retained intervals.

– Lags run backward through the retained intervals in calendar order, across days and overnight: $(t, i-j)$ denotes the j-th retained interval before (t,i), possibly on an earlier trading day.

– Exchange holidays and early-close days remain in the futures sample; the option-trading sample excludes early-close days.

– Exogenous series u: availability (missing-data) indicator, recorded before any filling,

$$a_{t,i} = \mathbf{1}\{u \text{ observed in interval } (t,i)\} \in \{0,1\}.$$

– Then $u_{t,i}$ is forward-filled (last observation carried forward), without a limit on staleness. Every interval with $a_{t,i} = 0$ is later imputed with the full-sample median (Section 3, step 4).

  - Diurnal adjustment (Andersen and Bollerslev, 1997). $\mathcal{D}_{t,i}$: the last 20 trading days $t' < t$ on which interval i is retained.

$$s^2_{t,i} = \frac{1}{|\mathcal{D}_{t,i}|} \sum_{t' \in \mathcal{D}_{t,i}} \mathrm{RV}_{t',i}, \qquad |\mathcal{D}_{t,i}| \geq 5.$$

  - $|\mathcal{D}_{t,i}| < 5$: $s^2_{t,i} = 1$, and the interval is excluded from evaluation.
  - Target, the diurnally adjusted realized volatility, and its winsorized version:

$$y_{t,i} = \sqrt{\mathrm{RV}_{t,i}/s^2_{t,i}}, \qquad \tilde{y}_{t,i} = \mathrm{clip}(y_{t,i},\ Q_{0.01}(\mathcal{Y}_{t,i}),\ Q_{0.99}(\mathcal{Y}_{t,i})),$$

$$\mathcal{Y}_{t,i} = \{y_{t,i-j} : j = 1, \ldots, 240\},$$

    with $Q_{0.01}(\cdot)$ and $Q_{0.99}(\cdot)$ the empirical 1st and 99th percentiles.
  - Winsorization window: the 240 preceding intervals of all times of day, excluding the current interval.

- **Samples** (Appendix Table 3).
  - Pooled intraday sample: 300,317 half-hour intervals from June 1998, with rolling estimation windows of $W = 24{,}000$ intervals (trees) and $W = 96{,}000$ intervals (linear models; Appendix B.2):

$$300{,}317 = 303{,}442 \text{ intervals on the grid} - 3{,}125 \text{ burn-in intervals.}$$

  - Closing-interval sample ($i = c$): 3,469 trading days, July 2010 to April 2024, with a rolling estimation window of $W = 2{,}000$ trading days.
  - 1,469 rolling-window forecasts from June 2018:

$$1{,}469 = 3{,}469 - 2{,}000.$$

  - Hyperparameter selection for the linear models on a pre-sample: closing-interval observations from 13 April 2004; validation samples from 6 January 2016 to 2 January 2020.
  - Evaluation span $\mathcal{T}_{\text{oos}}$: 3 January 2020 to 30 April 2024, the $T_{\text{oos}} = 1{,}082$ trading days on which every rolling forecast exists, less 6 early-close days. It starts with the pooled intraday linear forecasts, after the pre-sample above (Section 4.4).
  - Option-data subsample $\mathcal{T}_{\text{opt}} \subset \mathcal{T}_{\text{oos}}$, on which the straddle strategy is evaluated (Section 6):

$$T_{\text{opt}} = 866 \text{ trading days} = 872 \text{ SPXW expirations} - 6 \text{ early-close days.}$$

  - The 419 later expirations (413 full trading days) are held out.

3 Predictors

- **Eight groups of exogenous predictors, 47 series:** realized moments (7), liquidity (14), equal-weighted market (6), value-weighted market (6), sentiment (3), implied volatility (3), volatility demand (4), FOMC (4).

$$47 = 7 + 14 + 6 + 6 + 3 + 3 + 4 + 4.$$

– FOMC:

$$\mathbf{1}\{\text{announcement in } (t,i)\}, \quad \mathbf{1}\{\text{announcement on day } t\}, \quad 1/(1+n^{+}_{t,i}), \quad 1/(1+n^{-}_{t,i}).$$

– $n^{\pm}_{t,i}$: number of intervals to the next (+) and since the last (−) scheduled announcement.

- **Predictors built from each series.**
  – Trailing averages of the transformed series $\tilde{u}$ over horizons of k intervals, lagged one interval (HAR components; Corsi, 2009):

$$\bar{u}^{(k)}_{t,i} = \frac{1}{k}\sum_{j=1}^{k} \tilde{u}_{t,i-j}, \qquad k \in \mathcal{K},$$

$$\mathcal{K} = \begin{cases} \{1, 5, 25, 125, 625, 3125\} & \text{closing-interval model,} \\ \{2^0, 2^1, \ldots, 2^{11}\} \text{ (base 2, 12 horizons)} & \text{pooled intraday model.} \end{cases}$$

  – k counts intervals of the full intraday grid, overnight included; the closing-interval model uses the rows $i = c$.
  – The same averages of the availability indicator, $\bar{a}^{(k)}_{t,i}$.
  – Averages of an activity indicator, $\bar{\chi}^{(k)}_{t,i}$, for 7 zero-inflated series.
  – Names in tables and figures: the bar and the interval index are dropped ($i = c$ in the closing-interval model), so $u^{(k)}_t$ is series u averaged over k intervals; e.g. $\mathrm{VIX}^{(25)}_t$ is the transformed VIX averaged over the 25 intervals ending at 15:30 on day t. Series symbols: VIX, VIX3M, VVIX; constituent aggregates carry EW or VW, as in $\mathrm{OFI}^{\mathrm{EW}}$, $\mathrm{RV}^{\mathrm{VW}}$, $\mathrm{Spread}^{\mathrm{VW}}$; FOMC. Indicator averages: $\bar{a}^{(k)}_t(u)$, $\bar{\chi}^{(k)}_t(u)$.

- **HAR benchmark:** the HAR components of the target (6 columns) and 16 calendar dummies $\boldsymbol{D}_{t,i}$ (day of the week, monthly option expiration, quarterly "quadruple witching" expiration, month end, quarter end and others).

$$\bar{y}^{(k)}_{t,i} = \frac{1}{k}\sum_{j=1}^{k} \tilde{y}_{t,i-j}, \qquad 16 = \underbrace{5}_{\text{day of week}} + \underbrace{4}_{\text{time of day}} + \underbrace{7}_{\text{expiration, month, quarter}}.$$

  – At the closing interval, four of the 16 dummies (hour, overnight, opening and closing interval) are constant, and the quadruple-witching closing-auction dummy equals the quadruple-witching dummy, so the column screen below drops these five at every estimation of the closing-interval model.

- **Transformations, in order:**
  1. diurnal adjustment, dividing by a factor estimated on the same intraday interval ($\mathcal{D}_{t,i}$ as for $s^2_{t,i}$) and bounded relative to its own trailing median:

$$\tilde{u}_{t,i} = \frac{u_{t,i}}{b^{(u)}_{t,i}}, \qquad b^{(u)}_{t,i} = \begin{cases} \mathrm{clip}\big(\mathrm{mean}_{t' \in \mathcal{D}_{t,i}}\, u_{t',i},\ 0.5\,\rho_{t,i},\ 2\rho_{t,i}\big), & u \geq 0, \\ \max\big\{\mathrm{clip}\big(\mathrm{sd}_{t' \in \mathcal{D}_{t,i}}\, u_{t',i},\ 0.5\,\rho_{t,i},\ 2\rho_{t,i}\big),\ 0.1\,\bar{\sigma}^{(u)}\big\}, & u \text{ signed;} \end{cases}$$

     – $\rho_{t,i}$: median of the 250 most recent unbounded factors of interval i (at least 20 required).
     – $\bar{\sigma}^{(u)}$: full-sample median of the standard deviation on the same intraday interval.
     – Factor of a nonnegative series zero or undefined: $b^{(u)}_{t,i} = 1$.

– Not adjusted: VIX, VIX3M and VVIX, StockTwits sentiment, FOMC.

2. a variance-stabilizing transformation by series type, $\tilde{u}_{t,i} \leftarrow g(\tilde{u}_{t,i})$, the first matching rule applying:

$$g(v) = \begin{cases} \sqrt{v} & \text{squared returns, realized variance, turnover,} \\ & \text{bipower variation, effective spread,} \\ \operatorname{sign}(v)\,|v|^{1/2} & \text{autocovariance,} \\ v^{1/3} & \text{cubed returns,} \\ \operatorname{sign}(v)\,|v|^{1/4} & \text{fourth-power returns,} \\ v & \text{other signed series, absolute returns,} \\ \log \max(v, 10^{-12}) & \text{otherwise.} \end{cases}$$

– Signed series with no type rule whose standard-deviation divisor sits at its floor on at least 50% of intervals (volatility demand) instead take an inverse hyperbolic sine of the raw series:

$$\tilde{u}_{t,i} = \operatorname{asinh}(u_{t,i}/\kappa_{t,i}), \qquad \kappa_{t,i} = \max\{e_{t,i-1},\ 0.01 \operatorname{med}_{j\geq 1} e_{t,i-j}\}.$$

– $e_{t,i}$: exponentially weighted moving average of $|u|$ over intervals with $u \neq 0$, half-life 250 days.

3. winsorization at the rolling 1st and 99th percentiles, as for the target:

$$\tilde{u}_{t,i} \leftarrow \operatorname{clip}(\tilde{u}_{t,i},\ Q_{0.01}(\mathcal{U}_{t,i}),\ Q_{0.99}(\mathcal{U}_{t,i})), \qquad \mathcal{U}_{t,i} = \{\tilde{u}_{t,i-j} : j = 1, \ldots, 240\};$$

4. imputation of missing values by the full-sample median, kept alongside the availability indicator:

$$\tilde{u}_{t,i} \leftarrow \begin{cases} \tilde{u}_{t,i}, & a_{t,i} = 1, \\ \operatorname{med}\{\tilde{u}_{t',i'} : a_{t',i'} = 1\}, & a_{t,i} = 0; \end{cases}$$

5. activity indicators, for the 7 series whose full-sample mode accounts for at least 20% of intervals:

$$\chi_{t,i} = \mathbf{1}\{u_{t,i} \neq \operatorname{mode}(u)\};$$

6. trailing averages $\bar{u}^{(k)}_{t,i}$, $\bar{a}^{(k)}_{t,i}$, $\bar{\chi}^{(k)}_{t,i}$, $\bar{y}^{(k)}_{t,i}$;
7. robust standardization on the rolling estimation window (median and interquartile range, with floors), for the averages of series values only; written for the closing-interval model, whose rows t' are trading days ($i = c$ suppressed), with $\mathcal{W}_t = \{t - W, \ldots, t - 1\}$:

$$z^{(k)}_t = \frac{\bar{u}^{(k)}_t - \operatorname{med}_{t' \in \mathcal{W}_t} \bar{u}^{(k)}_{t'}}{\max\{\operatorname{IQR}^{(k)}_t,\ 0.01 \max_{t' \leq t} \operatorname{IQR}^{(k)}_{t'},\ \operatorname{IQR}^{*(k)}_t\}}, \qquad \operatorname{IQR}^{(k)}_t = \operatorname{IQR}\{\bar{u}^{(k)}_{t'} : t' \in \mathcal{W}_t\}.$$

– $\operatorname{IQR}^{*(k)}_t$: IQR of $\bar{u}^{(k)}_{t'}$, $t' < t$, over rows whose k-interval averaging window is fully observed.
– Column whose mode accounts for at least 20% of rows: $\operatorname{IQR}^{*(k)}_t$ also excludes rows at the mode.
– $\operatorname{IQR}^{*(k)}_t = 1$ until 8 such rows exist.
– $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$: not standardized. HAR components, calendar dummies: $\operatorname{IQR}^{*(k)}_t = 0$.
– $t < W$: the first window's statistics.

  - Pooled intraday model only (rows are intervals (t', i')): median and IQR over rows with $a_{t',i'} = 1$; fewer than 512 such rows give $z^{(k)} = 0$.
- **Column screening:** at every estimation, on the window $\mathcal{W}_t$, keep

$$\mathcal{J}_t = \{j : \boldsymbol{x}_{\cdot j} \text{ not constant}, \ \boldsymbol{x}_{\cdot j} \neq \boldsymbol{x}_{\cdot j'} \ \forall\, j' < j, \ j' \in \mathcal{J}_t\}, \qquad \boldsymbol{x}_{\cdot j} = (x_{t'j})_{t' \in \mathcal{W}_t}.$$

  - Equality is exact (bitwise); the first copy is kept.
  - The closing-interval model with the full predictor set keeps a median of 369 of its 628 columns.
- **Predictor sets:** the HAR benchmark has 22 columns; the full predictor set has 628.

$$628 = \underbrace{22}_{\bar{y}^{(k)},\ \boldsymbol{D}} + \underbrace{47 \times 6}_{\bar{u}^{(k)}} + \underbrace{47 \times 6}_{\bar{a}^{(k)}} + \underbrace{7 \times 6}_{\bar{\chi}^{(k)}}, \qquad 22 = 6 + 16.$$

  - The pooled intraday model's full predictor set has 1,144 columns: 43 series, 12 horizons, and 24 interactions of the HAR components with the opening- and closing-interval dummies.

$$1{,}144 = \underbrace{12 + 16}_{\bar{y}^{(k)},\ \boldsymbol{D}} + \underbrace{24}_{\bar{y}^{(k)} \times \text{open, close}} + 43 \times 12 + 43 \times 12 + 5 \times 12.$$

- **No look-ahead:** every statistic dated (t, i) uses only intervals before (t, i), so every closing-interval predictor is known at the 15:30 ET forecast origin: the trailing averages, $s^2_{t,i}$, the winsorization bounds, $\rho_{t,i}$, the standardization statistics and $\mathcal{J}_t$.
  - Full-sample elements, disclosed: the imputation median, the mode, $\bar{\sigma}^{(u)}$ and the assignment of series to the inverse hyperbolic sine.
  - The first window's standardization statistics use warm-up rows only.

4 Methods

4.1 Problem formulation

- **Rolling-window out-of-sample exercise,** re-estimated every period; every predictor is in the information set $\mathcal{F}_t$ at the 15:30 ET forecast origin. A period is a 30-minute intraday interval under the pooled intraday scheme (Section 4.3) and a trading day for the benchmark forecast (Section 4.4).
- **Notation:** $t = 1, \ldots, T$ indexes trading days and $i = 1, \ldots, M$ the 30-minute intraday intervals of day t, overnight included; $i = c$ is the closing interval, the closing half hour (15:30–16:00 ET).
  - $\mathrm{RV}_{t,i} = \sum_{k=1}^{N_{t,i}} r^2_{t,i,k}$: realized variance of interval (t, i), with $r_{t,i,k}$ its kth one-minute log return, $N_{t,i} \leq 30$ (Andersen et al., 2003).
  - The forecast target is the closing interval: $\mathrm{RV}_t \equiv \mathrm{RV}_{t,c}$.
  - $(t, i - j)$: the jth retained interval before (t, i), possibly on an earlier trading day.
  - $\mathcal{F}_t$: all data through 15:30 ET on day t; $\mathbb{E}_t[\cdot] \equiv \mathbb{E}[\cdot \mid \mathcal{F}_t]$ and $\mathrm{Var}_t(\cdot) \equiv \mathrm{Var}(\cdot \mid \mathcal{F}_t)$.
- **Diurnal adjustment** (Andersen and Bollerslev, 1997): the trailing mean of RV in the same interval, strictly before day t,

$$s^2_{t,i} = \frac{1}{|\mathcal{D}_{t,i}|} \sum_{t' \in \mathcal{D}_{t,i}} \mathrm{RV}_{t',i},$$

$$\mathcal{D}_{t,i} = \{\text{last } n_{\mathrm{d}} \text{ trading days } t' < t \text{ with interval } i\}, \qquad |\mathcal{D}_{t,i}| \geq \underline{n}_{\mathrm{d}}.$$

  - Intervals with $|\mathcal{D}_{t,i}| < \underline{n}_{\mathrm{d}}$: not scored. Values of n_{d} and $\underline{n}_{\mathrm{d}}$: Appendix B.1.

- **Target:** diurnally adjusted realized volatility (the square root stabilizes the variance), winsorized on that scale at rolling lower and upper empirical quantiles:

$$y_{t,i} = \sqrt{\mathrm{RV}_{t,i}/s_{t,i}^2}, \qquad \mathcal{Y}_{t,i} = \{y_{t,i-j} : j = 1, \ldots, n_{\mathrm{w}}\},$$
$$\tilde{y}_{t,i} = \mathrm{clip}(y_{t,i},\ \underline{Q}_{t,i},\ \overline{Q}_{t,i}).$$

  - $\underline{Q}_{t,i}$, $\overline{Q}_{t,i}$: lower and upper empirical quantiles of $\mathcal{Y}_{t,i}$, the previous n_{w} intervals, all intervals of the day pooled; clip truncates its first argument to the interval between the other two. Levels and n_{w}: Appendix B.1.
  - Models fit $\tilde{y}_{t,i}$; out-of-sample scores use the unwinsorized $\mathrm{RV}_{t,c}$.
  - $y_t \equiv y_{t,c}$ and $\tilde{y}_t \equiv \tilde{y}_{t,c}$: the closing interval's values.

- **Back-transform:** square and rescale, with no recalibration:

$$\hat{\sigma}_t^2 = \hat{y}_t^{\,2}\, s_{t,c}^2,$$

  where $\hat{y}_t$ is the forecast of y_t made with $\mathcal{F}_t$.
  - Jensen bias, downward, not corrected and disclosed: $\mathbb{E}_t[y_t^2] - (\mathbb{E}_t[y_t])^2 = \mathrm{Var}_t(y_t) \geq 0$.

- **Scoring loss:** QLIKE, robust to noise in the realized-variance proxy (Patton, 2011), evaluated at the actual (unwinsorized) realized variance:

$$\mathrm{QLIKE}(\mathrm{RV}, \hat{\sigma}^2) = \frac{\mathrm{RV}}{\hat{\sigma}^2} - \log\frac{\mathrm{RV}}{\hat{\sigma}^2} - 1, \qquad L_t = \mathrm{QLIKE}(\mathrm{RV}_t,\ \hat{\sigma}_t^2) = \mathrm{QLIKE}(y_t^2,\ \hat{y}_t^{\,2}).$$

  - $s_{t,c}^2$ cancels, since $\mathrm{RV}_t = y_t^2\, s_{t,c}^2$ with the unwinsorized y_t: the loss is scale-invariant.
  - Scores use RV_t, not the winsorized $\tilde{y}_t^{\,2} s_{t,c}^2$; the models still fit $\tilde{y}$.

- **Rolling estimation window** $\mathcal{W}_t = \{t-W, \ldots, t-1\}$: the last W periods before the forecast origin; $\mathcal{I}$ denotes the intervals used in estimation ($\mathcal{I}$ for each scheme: Section 4.3; values of W: Appendix B.2).

- **Hyperparameter-selection splits run forward in time** (Gu et al., 2020): an estimation sample, a gap of n_{gap} observations and a validation sample of n_{val} observations:

$$\mathcal{W}_t^{\mathrm{est}} = \{t - W, \ldots, t - n_{\mathrm{val}} - n_{\mathrm{gap}} - 1\},$$
$$\mathcal{W}_t^{\mathrm{gap}} = \{t - n_{\mathrm{val}} - n_{\mathrm{gap}}, \ldots, t - n_{\mathrm{val}} - 1\},$$
$$\mathcal{W}_t^{\mathrm{val}} = \{t - n_{\mathrm{val}}, \ldots, t - 1\}.$$

  Observation t never enters. Values of n_{gap} and n_{val}: Appendix B.1.

- **Estimation objective:** squared error on the scale of $\tilde{y}$. The forecast at origin t is estimated on the full window:

$$\hat{\boldsymbol{\theta}}_t = \arg\min_{\boldsymbol{\theta}} \sum_{t' \in \mathcal{W}_t} \left(\tilde{y}_{t'} - \hat{y}_{t'}(\boldsymbol{\theta})\right)^2.$$

  - t' indexes the observations of a window; $\boldsymbol{\theta}$: the model's parameters.
  - During hyperparameter selection only, each candidate configuration is estimated on the estimation sample $\mathcal{W}_t^{\mathrm{est}}$ in place of $\mathcal{W}_t$ and evaluated on the validation sample $\mathcal{W}_t^{\mathrm{val}}$.

4.2 Models

- **HAR and the HAR benchmark, by OLS** (Corsi, 2009):

$$\hat{y}_t = \beta_0 + \sum_{k\in\mathcal{K}} \beta^{(k)}\, \bar{y}_t^{(k)} + \boldsymbol{\delta}^\top \boldsymbol{D}_t, \qquad \bar{y}_t^{(k)} = \frac{1}{k}\sum_{j=1}^{k} \tilde{y}_{t,c-j}, \qquad \mathcal{K} = \{1, 5, 25, 125, 625, 3125\}.$$

  - $\bar{y}_t^{(k)}$: the HAR component at horizon k, the mean over the k half-hour intervals before the closing interval, overnight included.
  - $\boldsymbol{D}_t$: calendar dummies, coefficients $\boldsymbol{\delta}$; HAR alone sets $\boldsymbol{\delta} = \mathbf{0}$. The HAR benchmark ($\mathcal{X} = \text{HAR}$) is the HAR components and the calendar dummies.
  - $\boldsymbol{x}_t \in \mathbb{R}^p$: the full predictor set ($\mathcal{X} = \text{Full}$), the HAR benchmark's predictors plus, for each additional series u, the HAR components $\bar{u}_t^{(k)}$, $k \in \mathcal{K}$, and the averages of its availability and activity indicators (Section 3).

- **Ridge (Hoerl and Kennard, 1970), lasso (Tibshirani, 1996) and elastic net (Zou and Hastie, 2005),** written in the scikit-learn parametrization (the λ grids of Appendix B.3 are on this scale) and solved exactly as the window rolls:

$$\hat{\boldsymbol{\beta}}_t^{\text{ridge}} = \underset{\beta_0,\boldsymbol{\beta}}{\arg\min} \sum_{t'\in\mathcal{W}_t} (\tilde{y}_{t'} - \beta_0 - \boldsymbol{x}_{t'}^\top\boldsymbol{\beta})^2 + \lambda \sum_{j=1}^{p} \omega_j\, \beta_j^2,$$

$$\hat{\boldsymbol{\beta}}_t^{\text{enet}} = \underset{\beta_0,\boldsymbol{\beta}}{\arg\min} \frac{1}{2|\mathcal{W}_t|} \sum_{t'\in\mathcal{W}_t} (\tilde{y}_{t'} - \beta_0 - \boldsymbol{x}_{t'}^\top\boldsymbol{\beta})^2 + \lambda\alpha \sum_{j=1}^{p} \omega_j|\beta_j| + \frac{\lambda(1-\alpha)}{2} \sum_{j=1}^{p} \omega_j\, \beta_j^2,$$

  with $\lambda > 0$ the penalty strength, $\alpha \in [0, 1]$ the elastic-net mixing weight, and $\omega_j \geq 0$ the penalty weight on predictor j (intercept β_0 unpenalized, $\omega_0 = 0$):

  - **Block penalty** (block-penalized ridge, lasso and elastic net): the HAR benchmark is penalized 100 times more lightly than the additional predictors,

$$\omega_j = \begin{cases} \underline{\omega} = 0.01 & j \in \text{HAR benchmark}, \\ 1 & j \text{ an additional predictor}, \\ 0 & j = 0 \text{ (intercept)}. \end{cases}$$

  - HAR benchmark alone: $\omega_j = 1$ for every predictor, a single penalty.
  - Lasso: $\alpha = 1$. Elastic net: $\alpha = 0.5$.
  - Ridge: rank-one add and drop on $(\boldsymbol{X}_t^\top\boldsymbol{X}_t + \lambda\,\text{diag}(\boldsymbol{\omega}))^{-1}$, where $\boldsymbol{X}_t$ stacks the rows $(1, \boldsymbol{x}_{t'}^\top)$, $t' \in \mathcal{W}_t$, and $\boldsymbol{\omega} = (\omega_0, \ldots, \omega_p)^\top$.
  - Lasso, elastic net: exact online homotopy (Garrigues and El Ghaoui, 2008), one observation in, one out.
  - At every fit, only the screened columns $\mathcal{J}_t$ enter: predictors constant within the window or exact copies of another column are dropped; in a model estimated on the closing interval only, this always removes five of the 16 calendar dummies (Section 3).

- **Tree ensembles:** gradient boosting (LightGBM, XGBoost) and random forest.
  - **Gradient boosting** (Friedman, 2001), in LightGBM (Ke et al., 2017) and XGBoost

(Chen and Guestrin, 2016): loss $\frac{1}{2}(\tilde{y} - F)^2$, rounds $b = 1, \ldots, B$, learning rate η:

$$\begin{aligned} F_b &= F_{b-1} + \eta f_b, \\ f_b &= \arg\min_f \sum_{t' \in \mathcal{W}_t} \left[g_{t'} f(\boldsymbol{x}_{t'}) + \tfrac{1}{2} h_{t'} f(\boldsymbol{x}_{t'})^2 \right] \\ &\quad + \gamma^{\text{tree}} J + \lambda_1^{\text{tree}} \|\boldsymbol{w}\|_1 + \tfrac{1}{2} \lambda_2^{\text{tree}} \|\boldsymbol{w}\|_2^2. \end{aligned}$$

- $g_{t'} = F_{b-1}(\boldsymbol{x}_{t'}) - \tilde{y}_{t'}$ and $h_{t'} = 1$: first and second derivatives of the loss at F_{b-1}; f: a tree with J leaves and leaf weights $\boldsymbol{w}$; γ^{tree}: penalty per leaf; $\lambda_1^{\text{tree}}, \lambda_2^{\text{tree}}$: L1 and L2 penalties on $\boldsymbol{w}$.
- Leaf weight at $\lambda_1^{\text{tree}} = 0$: $w_j^* = -G_j/(H_j + \lambda_2^{\text{tree}})$, with $G_j = \sum_{t' \in \mathcal{R}_j} g_{t'}$ and $H_j = \sum_{t' \in \mathcal{R}_j} h_{t'} = n_j$ over the n_j observations $\mathcal{R}_j$ in leaf j.
- Leaf minimum $n_{\min}$: $n_j \geq n_{\min}$ (LightGBM's minimum leaf size); $H_j \geq n_{\min}$ (XGBoost's minimum child weight, an observation count since $h_{t'} = 1$).
- **Random forest** (Breiman, 2001): average of B trees f_b, each grown on a bootstrap sample,

$$\hat{y}_t = \frac{1}{B} \sum_{b=1}^{B} f_b(\boldsymbol{x}_t);$$

a random subset of the predictors is tried at each split.

4.3 Training schemes

- **Pooled intraday scheme** (the pooled intraday model, $\mathcal{I} = \{1, \ldots, M\}$), the paper's training scheme: estimated on all M 30-minute intervals of each day, overnight included; one observation per interval, so a period is an intraday interval and the window $\mathcal{W}_{t,i} = \{(t, i - W), \ldots, (t, i-1)\}$ holds the W intervals before (t, i).
- Only the forecasts of the closing interval $i = c$, made at the 15:30 ET forecast origin, are scored and traded.
- Values of W and M, re-estimation frequencies and the pooled predictor panel: Appendix B.2.
- A refinement that trains on the last 30 minutes of each trading day only is the subject of Section 6.4.

4.4 Forecast evaluation

- **Forecast and realization.** The variance forecast is the back-transformed fit, $\hat{\sigma}_t^2 = \hat{y}_t^2 \, s_{t,c}^2$ (Section 4.1), with no bias correction. The realization is the actual realized variance of the closing interval, $\text{RV}_t \equiv \text{RV}_{t,c}$; it is not winsorized (the winsorized $\tilde{y}_{t,c}$ enters estimation only).
- **Evaluation set.** Every forecast comes from a model re-estimated on its trailing window, so every forecast is out of sample. $\mathcal{T}_{\text{oos}}$ is the set of forecast origins, trading days, after the initial estimation window and after the pre-sample on which hyperparameters are selected, on which every forecast compared exists; early-close days are excluded, and $T_{\text{oos}} = |\mathcal{T}_{\text{oos}}|$ (its dates and T_{oos}: Experimental setup, Section 5.1).
- **Loss.** The daily QLIKE loss $L_t = \text{QLIKE}(\text{RV}_t, \hat{\sigma}_t^2)$ (Patton, 2011), defined in Section 4.1, and its average over $\mathcal{T}_{\text{oos}}$:

$$\overline{\text{QLIKE}} = \bar{L} = \frac{1}{T_{\text{oos}}} \sum_{t \in \mathcal{T}_{\text{oos}}} L_t.$$

- **Benchmark forecast.** The daily HAR regression (Corsi, 2009): OLS on the HAR benchmark, the HAR components and the calendar dummies (Section 4.2), estimated on the closing interval only, one observation per trading day, and re-estimated every trading day.
- **Relative loss.** For forecast A against forecast B (the benchmark forecast unless stated otherwise), the loss differential is $d_t = L_t^A - L_t^B$, and the percentage change in average QLIKE is

$$\Delta\% = 100\,\frac{\bar{d}}{\bar{L}^B}, \qquad \bar{d} = \frac{1}{T_{\text{oos}}}\sum_{t\in\mathcal{T}_{\text{oos}}} d_t.$$

  Its significance is tested with the Diebold–Mariano statistic of Section 4.5.
- **Squared-error and absolute-error loss,** in variance units:

$$\text{MSE} = \frac{1}{T_{\text{oos}}}\sum_{t\in\mathcal{T}_{\text{oos}}} \left(\text{RV}_t - \hat{\sigma}_t^2\right)^2, \qquad \text{MAE} = \frac{1}{T_{\text{oos}}}\sum_{t\in\mathcal{T}_{\text{oos}}} \left|\text{RV}_t - \hat{\sigma}_t^2\right|.$$

- **Out-of-sample** R^2 (Campbell and Thompson, 2008), with the historical mean of the realized variance, known at the forecast origin, as the reference forecast:

$$R^2_{\text{OOS}} = 1 - \frac{\sum_{t\in\mathcal{T}_{\text{oos}}}\left(\text{RV}_t - \hat{\sigma}_t^2\right)^2}{\sum_{t\in\mathcal{T}_{\text{oos}}}\left(\text{RV}_t - \overline{\text{RV}}_{t-1}\right)^2}, \qquad \overline{\text{RV}}_{t-1} = \frac{1}{|\{t' < t\}|}\sum_{t'<t}\text{RV}_{t'},$$

  the expanding mean over every earlier trading day (start in Section 5.1).
- **Calibration:** the ratio of average realized to average forecast variance, $\overline{\text{RV}}/\overline{\hat{\sigma}^2}$, and the frequency of under-prediction, $\frac{1}{T_{\text{oos}}}\sum_{t\in\mathcal{T}_{\text{oos}}} \mathbf{1}\{\text{RV}_t > \hat{\sigma}_t^2\}$.

4.5 Statistical tests

- **Diebold–Mariano test** (Diebold and Mariano, 1995) on the daily QLIKE loss differential of two forecasts A and B, with the Harvey–Leybourne–Newbold small-sample correction for one-step-ahead forecasts (Harvey et al., 1997):

$$d_t = L_t^A - L_t^B, \qquad \text{DM} = \sqrt{\frac{|\mathcal{O}|-1}{|\mathcal{O}|}}\,\frac{\bar{d}}{\sqrt{\hat{\Omega}/|\mathcal{O}|}}.$$

  - A, B: forecasts $f_{m,\mathcal{X},\mathcal{I}}$, indexed by the model m, the predictor set $\mathcal{X} \in \{\text{HAR}, \text{Full}\}$ and the intervals used in estimation $\mathcal{I} \in \{\{c\}, \{1,\ldots,M\}\}$.
  - $\mathcal{O}$: the days on which the two series are compared, numbered $t = 1,\ldots,|\mathcal{O}|$ in time order: $\mathcal{O} = \mathcal{T}_{\text{oos}}$, $|\mathcal{T}_{\text{oos}}| = T_{\text{oos}}$, for forecast accuracy and the option-data subsample $\mathcal{T}_{\text{opt}} \subset \mathcal{T}_{\text{oos}}$ for the straddle returns (Experimental setup, Section 5.1).
  - $\bar{d}$: mean loss differential over $\mathcal{O}$; two-sided p-value from the standard normal, $p = 2\{1 - \Phi(|\text{DM}|)\}$, with Φ the standard normal distribution function.
  - DM < 0 favors forecast A. "Significant" means significant at the 5% level ($p < 0.05$).
- **Newey–West variance** (Newey and West, 1987), Bartlett weights, lag truncation q_{NW}:

$$\hat{\Omega} = \hat{\gamma}_0 + 2\sum_{j=1}^{q_{\text{NW}}}\Big(1 - \frac{j}{q_{\text{NW}}+1}\Big)\hat{\gamma}_j, \qquad \hat{\gamma}_j = \frac{1}{|\mathcal{O}|}\sum_{t=j+1}^{|\mathcal{O}|}(d_t - \bar{d})(d_{t-j} - \bar{d}).$$

  - DM: $q_{\text{NW}} = \lfloor 4(|\mathcal{O}|/100)^{2/9} \rfloor$ (its value on $\mathcal{T}_{\text{oos}}$: Experimental setup, Section 5.1).

- **HAC t-statistic of a mean daily return** (Newey–West, no HLN correction), for a daily return series r_t, either a single strategy's return (a test of its mean return) or the difference between two strategies' returns (a test of the mean return differential):

$$t = \frac{\bar{r}}{\sqrt{\hat{\Omega}_r / |\mathcal{O}|}},$$

  with $\hat{\Omega}_r$ the Newey–West variance above computed on r_t in place of d_t and $q_{\mathrm{NW}} = \lfloor 1.5\,|\mathcal{O}|^{1/3} \rfloor$ (its value on $\mathcal{T}_{\mathrm{opt}}$: Experimental setup, Section 5.1).
- **Model confidence set** (Hansen et al., 2011), **superior predictive ability** (Hansen, 2005) and **StepM** (Romano and Wolf, 2005) for the comparison of predictor sets.
- **Point estimates only;** no bootstrap confidence intervals in the main tables.

5 Experiments and results

5.1 Experimental setup

- **Evaluation span** $\mathcal{T}_{\mathrm{oos}}$ (Section 4.4): the rolling forecasts from 3 January 2020 to 30 April 2024, $T_{\mathrm{oos}} = |\mathcal{T}_{\mathrm{oos}}| = 1{,}082$ trading days, every trading day on which all forecasts exist (6 early-close days dropped). It starts with the pooled intraday linear forecasts, after the pre-sample on which the linear models' penalties are selected (6 January 2016 to 2 January 2020).
- **Option-data subsample** $\mathcal{T}_{\mathrm{opt}} \subset \mathcal{T}_{\mathrm{oos}}$: the $T_{\mathrm{opt}} = 866$ days with a same-day SPXW expiration (Section 2), on which the straddle economics of Section 6 are evaluated; the other 216 days enter forecast accuracy only.
- **Training scheme:** pooled intraday, all $M = 48$ intervals of each day; $W = 24{,}000$ intervals for the trees and 96,000 for the linear models. On the full predictor set, the pooled LightGBM and XGBoost are the earlier draft's specification and the pooled random forest is preliminary (Table 1).
- **Benchmark forecast:** the daily HAR regression, $W = 2{,}000$ trading days; the models trained the same way, on the last 30 minutes of each day, are in Section 6.4.
- **Re-estimated every period** (Section 4.3); hyperparameters are fixed or selected causally, on a pre-sample or on validation samples that precede the forecast origin.
- **Loss and tests:** QLIKE against the actual realized variance (Section 4.4); Diebold–Mariano tests with the Harvey–Leybourne–Newbold correction ($q_{\mathrm{NW}} = 6$ on $\mathcal{T}_{\mathrm{oos}}$) and HAC t-statistics of straddle returns ($q_{\mathrm{NW}} = 14$ on $\mathcal{T}_{\mathrm{opt}}$), each with a Newey–West long-run variance (Section 4.5).

Full implementation details are in Appendix B.

5.2 Out-of-sample accuracy

- **Overview:** Table 1 reports the out-of-sample MSE, average QLIKE and R^2_{OOS} of every pooled intraday model, on the HAR benchmark and on all features, over the 1,082 trading days of the evaluation span $\mathcal{T}_{\mathrm{oos}}$. The same statistics for the models trained on the last 30 minutes, whose OLS on the HAR benchmark is the benchmark forecast, are in Table 2 (Section 6.4).
- R^2_{OOS} follows Campbell and Thompson (2008): the benchmark is the historical mean of the closing-half-hour realized variance from 5 January 1998 through the day before the forecast, so it uses no out-of-sample information.

5.3 The full predictor set versus the HAR benchmark

- **Paired comparison** within each model class: the same pooled intraday model m ($\mathcal{I} = \{1, \ldots, M\}$) on the two predictor sets, $d_t = L_t^{\text{Full}} - L_t^{\text{HAR}}$, with $L_t^{\mathcal{X}}$ the loss of the forecast $f_{m,\mathcal{X},\{1,\ldots,M\}}$:

$$\Delta\% = 100\Big(\frac{\bar{L}^{\text{Full}}}{\bar{L}^{\text{HAR}}} - 1\Big) = 100\,\frac{\bar{d}}{\bar{L}^{\text{HAR}}}, \qquad \text{DM on } d_t,\ t \in \mathcal{T}_{\text{oos}},\ q_{\text{NW}} = 6.$$

- **The full predictor set lowers average QLIKE in every model class** ($\Delta\% < 0$ in all 6), 4 of them significantly at the 5% level (Table 1):
  - trees, the largest reductions: $\Delta\% = -13.0$, -12.0 and -11.9 for LightGBM, XGBoost and the random forest (DM $= -4.39$, -4.29, -4.75), each significant;
  - linear models: elastic net $\Delta\% = -5.6$ (DM $= -2.10$, significant), ridge -5.0 (DM $= -1.77$, $p = 0.077$), lasso -3.0 (DM $= -1.13$, $p = 0.26$).
- **Economic value of the reductions:** on the option-data subsample $\mathcal{T}_{\text{opt}}$, the full predictor set raises the annualized Sharpe ratio of the sign strategy at mid-quotes in all 6 classes, by $+0.30$ to $+0.61$, none significantly (HAC t-statistic on the daily return differential, Newey–West, $q_{\text{NW}} = 14$; Section 6.3).
- **Squared-error loss does not confirm the reductions:** the full predictor set raises the MSE in 3 of the 6 classes and lowers it in the others, none significantly (DM on the squared error).
  - Each pooled forecast's five largest squared errors on the full predictor set fall in March 2020 and carry 72–83% of its $\sum_{t\in\mathcal{T}_{\text{oos}}}(\text{RV}_t - \hat{\sigma}_t^2)^2$, but only 1–4% of its total QLIKE loss.

5.4 Model classes

- **The linear models are the most accurate pooled intraday forecasts on each predictor set** (Table 1):
  - full predictor set: average QLIKE 0.1078–0.1101 for the linear models (elastic net 0.1078, ridge 0.1086, lasso 0.1101, the three most accurate of the 12 pooled forecasts) and 0.1126–0.1146 for the trees;
  - HAR benchmark: 0.1135–0.1143 for the linear models and 0.1279–0.1301 for the trees, the least accurate pooled forecasts.
- **Every pooled intraday forecast has a higher average QLIKE than the benchmark forecast** ($\overline{\text{QLIKE}} = 0.1033$; $\Delta\%$ and DM against it):
  - full predictor set: linear models $+4.4$ to $+6.7\%$, none significant at the 5% level; trees $+9.0$ to $+11.0\%$, each significant at the 5% level;
  - HAR benchmark: $+10.0$ to $+26.0\%$, each significant at the 1% level;
  - the full predictor set narrows the difference in every model class;
  - the benchmark forecast is trained on the last 30 minutes of each day; Section 6.4 trains every model class that way.

5.5 Variable importance

- **Models:** the closing-interval models of record on the full predictor set ($p = 628$), trained on the last 30 minutes of each trading day (Section 6.4), not on the pooled intraday scheme.

Table 1: Forecast accuracy of the pooled intraday models

	Pooled intraday		
Model and predictor set	MSE	QLIKE	R^2_{OOS} (%)
Panel A: Linear models			
OLS			
HAR benchmark[a]	4.82	0.1291	74.7
Ridge			
HAR benchmark	4.74	0.1143	75.1
All features	4.79	0.1086	74.9
Lasso			
HAR benchmark	4.91	0.1135	74.2
All features	4.70	0.1101	75.4
Elastic net			
HAR benchmark	4.94	0.1142	74.1
All features	4.72	0.1078	75.3
Panel B: Tree ensembles			
LightGBM			
HAR benchmark	3.98	0.1296	79.1
All features[a]	4.13	0.1128	78.3
XGBoost			
HAR benchmark	3.85	0.1279	79.8
All features[a]	4.01	0.1126	79.0
Random forest			
HAR benchmark	3.92	0.1301	79.4
All features[†]	3.75	0.1146	80.3

Notes: Rolling forecasts from 3 January 2020 to 30 April 2024, every forecast made with a model estimated on its trailing window only, so every forecast is out of sample; each is scored against the actual realized variance of the closing half hour, RV_t, on the $T_{\text{oos}} = 1{,}082$ trading days on which every forecast of both training schemes exists, early closes dropped (Section 4.4). The span starts with the pooled intraday linear forecasts, on the first trading day after the pre-sample on which the penalties were selected (6 January 2016 to 2 January 2020). MSE is the mean squared error $T_{\text{oos}}^{-1}\sum_t(\text{RV}_t - \hat{\sigma}_t^2)^2$, multiplied by 10^{10}, and QLIKE the mean QLIKE loss. $R^2_{\text{OOS}} = 1 - \sum_t(\text{RV}_t - \hat{\sigma}_t^2)^2 / \sum_t(\text{RV}_t - \overline{\text{RV}}_{t-1})^2$, in percent, where the benchmark $\overline{\text{RV}}_{t-1}$ is the historical mean of the closing-half-hour realized variance over all trading days from 5 January 1998 to day $t-1$ (Campbell and Thompson, 2008). All features is the full predictor set of Section 3. The same days for the models trained on the last 30 minutes are in Table 2 (Section 6.4).

[a] The earlier draft's pooled intraday specification.

[†] Preliminary: the closing-interval forecasts of the pooled intraday random forest on all features, from an exercise still in progress.

  - Lasso and ridge with the penalty strength λ selected on a pre-sample: $\lambda = 10^{-2}$ and 10^{3} (scikit-learn parametrization). Penalty weight ω_j, the factor on predictor j's penalty term: $\underline{\omega} = 0.01$ on the HAR benchmark block, 1 on the exogenous predictors. Re-estimated every trading day; the scheduled re-selection of λ never replaced the pre-sample choice.
  - Not the earlier study's single-penalty specification (one λ for every predictor, re-selected every 250 trading days): the paper does not use it.
  - Tree ensembles at fixed hyperparameters, the best entries of the development-sample search (LightGBM and XGBoost: the top Optuna trial; random forest: trial 255); constant and duplicate predictors dropped in each estimation window.
- **Notation.**
  - Re-estimation $k = 1, \ldots, K$, $K = 147$, one every 10 trading days. Re-estimation k uses the window $\mathcal{W}_{t_k}$, the 2,000 trading days before its first forecast day t_k, and forecasts its out-of-sample block $\mathcal{O}_k$, the trading days from t_k to the next re-estimation (≤ 10). The blocks make up this section's forecast days $\mathcal{O} = \bigcup_k \mathcal{O}_k$: $|\mathcal{O}| = 1{,}469$ trading days, 25 June 2018 to 30 April 2024.
  - Predictors $\boldsymbol{x}_t \in \mathbb{R}^p$; forecast $\hat{y}_t = \hat{f}_k(\boldsymbol{x}_t)$, $t \in \mathcal{O}_k$, on the scale of the target y_t, the diurnally adjusted realized volatility of the closing half hour.
  - Predictor group $A \subseteq \{1, \ldots, p\}$: all predictors built from one series (its averages at every horizon, its missing-data and activity indicators). 49 groups: 47 exogenous series in 8 categories (Section 3), the HAR components, the calendar dummies.
  - Linear models: $\hat{\boldsymbol{\beta}}_k \in \mathbb{R}^p$, the coefficient estimates behind the forecast for t_k; $\hat{\boldsymbol{\beta}}_{k,A}$, its subvector on group A.
  - Trees: SHAP values $\phi_{t,j}$ (Lundberg and Lee, 2017) and SHAP interaction values $\phi_{t,jl}$ (Lundberg et al., 2020); they add up to the forecast,

$$\hat{y}_t = \phi_0 + \sum_{j=1}^{p} \phi_{t,j}, \qquad \phi_{t,j} = \sum_{l=1}^{p} \phi_{t,jl}, \qquad \phi_{t,A} = \sum_{j \in A} \phi_{t,j}.$$

  - Path-dependent TreeSHAP (Lundberg et al., 2020): the training observations per node (node covers) stand in for the background sample.
  - Number triples below: LightGBM, XGBoost, random forest.
- **Lasso, aggregate:** the selection count of a group and of a predictor, the number of the K re-estimations with a nonzero coefficient,

$$K_A = \sum_{k=1}^{K} \mathbf{1}\{\exists j \in A : \ \hat{\beta}_{k,j} \neq 0\}, \qquad K_j = \sum_{k=1}^{K} \mathbf{1}\{\hat{\beta}_{k,j} \neq 0\};$$

  K_A/K is the selection frequency, the share of estimation windows with a nonzero coefficient.
  - Per re-estimation: 30 nonzero coefficients (median; range 27–35), of which 14 on exogenous predictors.
  - Predictors: 60 of 628 selected at least once, 29 in at least half of the re-estimations.
  - Exogenous series: 27 of 47 selected at least once; 11 in at least half of the re-estimations; 3 in every re-estimation (r, RAV^{VW}, AC).
  - HAR benchmark block: selected in every re-estimation ($\omega_j = 0.01$); 88 % of $\sum_j |\hat{\beta}_{k,j}|$ (median re-estimation).

  - Random-probe benchmark: the continuous probe, an i.i.d. uniform noise predictor (defined under Known issues), is selected in 2.0 % of its fits; 54 predictors, in 23 exogenous series, are selected more often.

- **Lasso, time-varying:** the magnitude and the selection indicator of each group at each re-estimation,

$$\|\hat{\boldsymbol{\beta}}_{k,A}\|_1 = \sum_{j \in A} |\hat{\beta}_{k,j}|, \qquad \mathbf{1}\{\|\hat{\boldsymbol{\beta}}_{k,A}\|_1 > 0\}$$

  Rank stability across re-estimations: Kendall's coefficient of concordance W of the rankings by $\|\hat{\boldsymbol{\beta}}_{k,A}\|_1$ at each re-estimation ($W = 1$: the same ranking at every re-estimation; formula below), $W = 0.74$.
- **Ridge, aggregate:** the group's share of the absolute coefficient mass,

$$S_A = \frac{\sum_k \sum_{j \in A} |\hat{\beta}_{k,j}|}{\sum_k \sum_{j=1}^{p} |\hat{\beta}_{k,j}|}.$$

  - Scale check: only the averages of series values are robust-scaled.
    * Averages of series values (282 predictors): standard deviation over the estimation window 0.84 (median; 5th–95th percentiles: 0.03–1.76), interquartile range 0.94.
    * Averaged missing-data and activity indicators (324): standard deviation 0.022; calendar dummies 0.39; HAR components 1.07.
    * So the raw $|\hat{\beta}_{k,j}|$ are in different units. Check with the group's share of the absolute standardized coefficients,

$$S_A^{\mathrm{sd}} = \frac{\sum_k \sum_{j \in A} |\hat{\beta}_{k,j}| \, \mathrm{sd}_{k,j}}{\sum_k \sum_{j=1}^{p} |\hat{\beta}_{k,j}| \, \mathrm{sd}_{k,j}},$$

      with $\mathrm{sd}_{k,j}$ the standard deviation of predictor j over $\mathcal{W}_{t_k}$; Spearman rank correlation of S_A^{sd} with S_A across groups 0.85.
    * Leading groups on that scale: HAR components 30.6 %, calendar dummies 9.3 %, r 4.5 %.
  - Leading groups: calendar dummies 31.0 %, HAR components 22.8 %, r 3.6 %; HAR benchmark block 54 % (median re-estimation).
  - Ridge sets no coefficient to zero: all 49 groups have $K_A = K$, so the selection frequency does not discriminate.
  - Random-probe benchmark: 107 predictors, in 38 exogenous series, have $|\hat{\beta}_{k,j}|$ above the continuous probe's (robust-scaled) in most probe fits.
- **Ridge, time-varying:** the share at re-estimation k, $S_{k,A} = \|\hat{\boldsymbol{\beta}}_{k,A}\|_1 / \sum_{A'} \|\hat{\boldsymbol{\beta}}_{k,A'}\|_1$, A' over all groups (mean rank by $\|\hat{\boldsymbol{\beta}}_{k,A}\|_1$); Kendall's $W = 0.85$.
- **Trees, ranking at each re-estimation:** the group's mean absolute SHAP value on the out-of-sample block of re-estimation k, then its rank,

$$I_{k,A} = \frac{1}{|\mathcal{O}_k|} \sum_{t \in \mathcal{O}_k} |\phi_{t,A}|, \qquad r_{k,A} = 1 + \#\{A' : I_{k,A'} > I_{k,A}\} \quad \text{(ties averaged)}.$$

- **Trees, global ranking:** aggregate the K rankings over time,

$$\bar{r}_A = \frac{1}{K} \sum_{k=1}^{K} r_{k,A}, \qquad \mathrm{BC}_A = \sum_{k=1}^{K} \max(0,\ 11 - r_{k,A}).$$

  - BC_A: truncated Borda count (a full Borda count is affine in $\bar{r}_A$).
  - Kendall's coefficient of concordance (Kendall and Babington Smith, 1939) over the K rankings of the n groups, with rank sums $R_A = \sum_k r_{k,A}$ and their mean $\bar{R}$:

$$W = \frac{12 \sum_A (R_A - \bar{R})^2}{K^2(n^3 - n) - K \sum_k \sum_{g \in \mathcal{E}_k} (|g|^3 - |g|)},$$

    $\mathcal{E}_k$ the sets of tied (equally ranked) groups at re-estimation k; $W = 1$: the same ranking at every re-estimation.
  - LightGBM: HAR components ($\bar{r}_A = 1.0$), r (2.9), RAV (4.8); Kendall's $W = 0.57$.
  - XGBoost: HAR components (1.0), r (3.0), RAV (3.9); Kendall's $W = 0.60$.
  - Random forest: HAR components (1.0), RAV (2.6), r (4.0); Kendall's $W = 0.62$.
  - Mean rank vs truncated Borda count, Spearman rank correlation over groups: 0.90, 0.87, 0.92.
  - Aggregate of the ranks vs rank of the aggregate, the mean absolute SHAP value over $\mathcal{O}$, $\bar{I}_A = |\mathcal{O}|^{-1} \sum_{t \in \mathcal{O}} |\phi_{t,A}|$: 0.96, 0.97, 0.98.
  - Predictor level ($n = 628$): Kendall's $W = 0.91$, 0.90, 0.93.
  - Agreement of $\bar{r}_A$ across models (Spearman rank correlation): trees 0.83–0.96; tree vs linear 0.17–0.55; lasso vs ridge 0.47.
- **Interaction effects, global:** the mean absolute SHAP interaction value of two groups over $\mathcal{O}$,

$$\phi_{t,AB} = \sum_{j \in A} \sum_{l \in B} (\phi_{t,jl} + \phi_{t,lj}), \qquad \bar{I}_{AB} = \frac{1}{|\mathcal{O}|} \sum_{t \in \mathcal{O}} |\phi_{t,AB}|, \quad A \neq B;$$

  for two predictors, $|\phi_{t,jl} + \phi_{t,lj}|$, $j \neq l$.
  - $\phi_{t,jl} = \frac{1}{2}(\phi_{t,l} \mid j \text{ present} - \phi_{t,l} \mid j \text{ absent})$, $j \neq l$; $\phi_{t,jj} = \phi_{t,j} - \sum_{l \neq j} \phi_{t,jl}$.
  - Top pair: LightGBM HAR components × RAV (12.6); XGBoost HAR components × RAV (15.2); random forest HAR components × RAV (23.8); $\times 10^{-3}$, in units of y_t.
  - Scale: largest $\bar{I}_A$ 195, 192, 224 ($\times 10^{-3}$); all series-pair $\bar{I}_{AB}$ summed, as % of $\sum_A \bar{I}_A$: 90, 74, 72.
  - Within a group ($\sum_{j \neq l \in A} \phi_{t,jl}$): largest HAR components (18.7), HAR components (19.1), HAR components (34.3); $\times 10^{-3}$.
  - Noise floor: the continuous probe's largest $\bar{I}$ with any group, median over probe fits: 0.49, 0.41, 0.50 ($\times 10^{-3}$); series pairs above it: 116, 115, 52 of 1,176.
- **Interaction effects, cross-check:** the H-statistic of Friedman and Popescu (2008) on each model's top 10 series pairs, from the partial dependence functions PD_S, $S \in \{A, B, AB\}$, $AB \equiv A \cup B$:

$$H^2_{AB} = \frac{\sum_t [\mathrm{PD}_{AB}(\boldsymbol{x}_t) - \mathrm{PD}_A(\boldsymbol{x}_t) - \mathrm{PD}_B(\boldsymbol{x}_t)]^2}{\sum_t \mathrm{PD}_{AB}(\boldsymbol{x}_t)^2}, \qquad \mathrm{PD}_S(\boldsymbol{x}) = \frac{1}{|\mathcal{H}_k|} \sum_{t' \in \mathcal{H}_k} \hat{f}_k(\boldsymbol{x}_S, \boldsymbol{x}_{t',-S}),$$

  $t \in \mathcal{H}_k$; each PD_S centered over $\mathcal{H}_k$.
  - Points: $\mathcal{H}_k$, the 100 trading days from t_k on, both evaluation points and background sample, at 15 re-estimations (every tenth); sums pooled over re-estimations.
  - The denominator contains the HAR components' own partial dependence whenever they are in the pair. A second normalization, by the smaller single-group partial dependence:

$$H^2_{\min,AB} = \frac{\sum_t [\mathrm{PD}_{AB}(\boldsymbol{x}_t) - \mathrm{PD}_A(\boldsymbol{x}_t) - \mathrm{PD}_B(\boldsymbol{x}_t)]^2}{\min\{\sum_t \mathrm{PD}_A(\boldsymbol{x}_t)^2, \ \sum_t \mathrm{PD}_B(\boldsymbol{x}_t)^2\}}.$$

  - Top pair, H^2 (%): 0.33, 0.37, 0.85; over the top 10: 0.02–0.33, 0.02–0.68, 0.02–0.85.
  - Top pair, $H^2_{\min}$: 0.19, 0.39, 1.50; over the top 10: 0.05–0.54, 0.08–0.80, 0.17–5.46.
  - Top predictor pair, H^2 (%) and $H^2_{\min}$: LightGBM 0.36, 0.02; XGBoost 0.39, 0.02; random forest 1.16, 0.06.
- **Interaction effects, time-varying:** $\phi_{t,AB}$ on the out-of-sample block of each re-estimation, ranked at each re-estimation and aggregated as the SHAP values are,

$$I_{k,AB} = \frac{1}{|\mathcal{O}_k|} \sum_{t \in \mathcal{O}_k} |\phi_{t,AB}| \;\rightarrow\; r_{k,AB} \;\rightarrow\; \bar{r}_{AB},\ W.$$

  - Kendall's W over each model's top 50 pairs: 0.45, 0.47, 0.57.
  - Mean rank vs $\bar{I}_{AB}$ (Spearman rank correlation, all pairs): 0.99, 0.99, 1.00.
- **Known issues, reconciled:**
  - Mean decrease in impurity (MDI) favors predictors with many distinct values (Strobl et al., 2007): MDI is not used; SHAP values and coefficients only.
  - Random probes, at every third re-estimation (49 fits): three i.i.d. noise predictors, continuous (U(0, 1)), discrete uniform on 20 levels, and binary (Bernoulli, probability one half).
    * Lasso: the probes are selected in 1, 0, 0 of 49 fits (robust-scaled, $\omega_j = 1$).
    * Ridge: 107 predictors have $|\hat{\beta}_{k,j}|$ above the continuous probe's in most probe fits.
    * Trees: the continuous probe's median rank among 631 predictors by mean $|\phi_{t,j}|$: 83, 110, 97; exogenous series with a predictor above it in most probe fits: 27, 35, 29.
  - Dilution across correlated predictors: $\phi_{t,A}$ and $\|\hat{\boldsymbol{\beta}}_{k,A}\|_1$ sum over the group's predictors, so credit split among a series' averages at different horizons stays inside the group.
  - The earlier study's single-penalty lasso (λ re-selected): median 23 nonzero coefficients per re-estimation; 18 exogenous series selected in at least half of the re-estimations; Spearman rank correlation of its K_A with that of the lasso of record 0.70.
  - Reproduction: re-running the estimation reproduces the archived LightGBM models exactly; XGBoost and the random forest differ across library versions (maximum absolute forecast difference 0.17 and 0.11, in units of y_t). SHAP values: from the archived models; interaction values: from the reproduced ones (Spearman rank correlation of $\bar{r}_A$, reproduced vs archived: 1.000, 0.993, 0.996).
- **Compute** (one laptop, 11 processes):
  - Reproducing the tree models and computing the SHAP interaction values: 133 minutes. Per re-estimation: random forest estimation 212 s, its SHAP and interaction values 182 s; LightGBM 0.9 s, XGBoost 0.4 s.
  - Linear coefficient paths and probes: 7 minutes; H^2: 7 minutes.

6 Economic value of the forecasts

6.1 The straddle strategy

- **Position:** SPXW options that expire the same day (0DTE). With the index at S_t at the 15:30 ET forecast origin, the call and the put at the nearest strikes on either side,

$$K_C = \min\{K \geq S_t\}, \qquad K_P = \max\{K \leq S_t\},$$

over strikes with live quotes. If $K_C = K_P = S_t$ the package is an at-the-money straddle; otherwise it is a strangle on the two strikes adjacent to S_t. We call both the straddle: a position that is approximately delta-neutral at entry, entered at 15:30 ET and held, not hedged, to expiration.

- **Entry prices:** at the mid-quote, P_t, and at the quoted bid and ask of the package, P_t^b and P_t^a, from the legs' bid quotes b and ask quotes a at 15:30 ET:

$$P_t = \tfrac{1}{2}(b_t^C + a_t^C) + \tfrac{1}{2}(b_t^P + a_t^P), \qquad P_t^b = b_t^C + b_t^P, \qquad P_t^a = a_t^C + a_t^P.$$

- **Settlement** in cash at the official close S_t^{close}, with no delta hedging; R_t is the straddle's simple return on the premium paid at the mid-quote:

$$X_t = \left(S_t^{\text{close}} - K_C\right)^+ + \left(K_P - S_t^{\text{close}}\right)^+, \qquad R_t = \frac{X_t}{P_t} - 1 \;\geq\; -1.$$

- **Implied variance:** IV_t^C and IV_t^P are the legs' implied volatilities at 15:30 ET, quoted per hour; the implied volatility of the closing half hour is

$$\text{IV}_t = \frac{\text{IV}_t^C + \text{IV}_t^P}{2\sqrt{2}},$$

and IV_t^2 is the implied variance to expiration.

- **Forecast variance risk premium:** the variance risk premium of the closing half hour is $\text{IV}_t^2 - \mathbb{E}_t[\text{RV}_t]$ (Carr and Wu, 2009; Bollerslev et al., 2009), and its forecast is $\text{IV}_t^2 - \hat{\sigma}_t^2$, with $\hat{\sigma}_t^2$ the variance forecast of Section 4.1. Every strategy below conditions on the forecast only through the gap $\hat{\sigma}_t^2 - \text{IV}_t^2$: it buys when the forecast exceeds implied variance.
- **Strategies:** position q_t in straddles ($q_t > 0$ long, $q_t < 0$ short) and daily return $r_t = q_t R_t$. The forecast enters only through q_t, so the comparison across strategies measures the economic value of the forecast (Fleming et al., 2001):
  - the unconditional short-straddle benchmark, a static short-variance position: $q_t \equiv -1$;
  - sign strategy: $q_t = \text{sign}(\hat{\sigma}_t^2 - \text{IV}_t^2)$, with $q_t = -1$ on ties;
  - robust clip: $q_t = \text{clip}\big((\hat{\sigma}_t^2 - \text{IV}_t^2)/\text{MAD}_t, -1, 1\big)$, with the scale $\text{MAD}_t = \text{median}_{t'<t}|\hat{\sigma}_{t'}^2 - \text{IV}_{t'}^2|$, the median absolute gap over the earlier days of $\mathcal{T}_{\text{opt}}$;
  - full long, scaled short: $q_t = 1$ if $\hat{\sigma}_t^2 > \text{IV}_t^2$, else $q_t = \max\big((\hat{\sigma}_t^2 - \text{IV}_t^2)/\text{MAD}_t, -1\big)$;
  - rank of the absolute gap, over the $t-1$ earlier days $t' < t$ of $\mathcal{T}_{\text{opt}}$:

$$q_t = \text{sign}(\hat{\sigma}_t^2 - \text{IV}_t^2) \cdot \frac{1}{t-1} \sum_{t'<t} \mathbf{1}\{|\hat{\sigma}_{t'}^2 - \text{IV}_{t'}^2| \leq |\hat{\sigma}_t^2 - \text{IV}_t^2|\}.$$

- **Sizing sample:** q_t uses days $t' < t$ only. The first 252 days of $\mathcal{T}_{\text{opt}}$ build the scale MAD_t, so every sizing rule, the sign strategy included, is evaluated on the remaining 614 trading days, $t > 252$.
- **Execution:** at the mid-quote, $r_t = q_t R_t$. At quoted bid/ask prices a purchase pays the ask and a sale receives the bid; the spread is paid at entry only, since settlement is in cash:

$$r_t^{\text{ba}} = \begin{cases} |q_t|\,(X_t/P_t^a - 1), & q_t > 0, \\ |q_t|\,(1 - X_t/P_t^b), & q_t < 0. \end{cases}$$

- **Transaction costs:** trading at the quoted bid or ask instead of the mid-quote costs a median 2.9% of the premium. No other transaction costs are deducted.

6.2 Performance measures

- **Annualized Sharpe ratio,** from daily returns, at mid-quotes and at quoted bid/ask prices, over the days $t \in \mathcal{T}_{\text{opt}}$ of the option-data subsample, $|\mathcal{T}_{\text{opt}}| = T_{\text{opt}}$ (the 614 days with $t > 252$ for the sizing rules):

$$\text{SR} = \sqrt{252}\,\frac{\bar{r}}{\text{sd}(r)}, \qquad \text{sd}(r)^2 = \frac{1}{T_{\text{opt}} - 1}\sum_{t\in\mathcal{T}_{\text{opt}}} \big(r_t - \bar{r}\big)^2.$$

  The risk-free return over the half-hour holding period is negligible and is not subtracted.
- **Mean return and its HAC t-statistic:** $t = \bar{r}/\sqrt{\hat{\Omega}_r/T_{\text{opt}}}$, with $\hat{\Omega}_r$ the Newey–West long-run variance of Section 4.5 applied to r_t and lag truncation $q_{\text{NW}} = \lfloor 1.5\,T_{\text{opt}}^{1/3} \rfloor = 14$.
- **Hit rate** $\frac{1}{T_{\text{opt}}}\sum_{t\in\mathcal{T}_{\text{opt}}} \mathbf{1}\{r_t > 0\}$; number of long positions $\#\{t : q_t > 0\}$.
- **Gain over the short-straddle benchmark** (or over any benchmark strategy B), with the daily return differential $\Delta_t = r_t^A - r_t^B$:

$$\Delta\text{SR} = \text{SR}(r^A) - \text{SR}(r^B), \qquad t_{\text{HAC}} = \frac{\bar{\Delta}}{\sqrt{\hat{\Omega}_\Delta/T_{\text{opt}}}},$$

  the HAC t-statistic on the daily return differential (Newey–West, $q_{\text{NW}} = 14$). It tests the mean return differential, not the equality of the two Sharpe ratios (Ledoit and Wolf, 2008).
- **Elementary score** of the Murphy diagram (Ehm et al., 2016) at threshold θ on the scaled values, evaluated at the implied variance IV_t^2, that is, at $\theta = 1$:

$$\text{ES}_{\theta,t} = \big|\varrho_t - \theta\big|\,\mathbf{1}\{\min(\hat{\varrho}_t, \varrho_t) \le \theta < \max(\hat{\varrho}_t, \varrho_t)\}, \qquad \hat{\varrho}_t = \frac{\hat{\sigma}_t^2}{\text{IV}_t^2}, \quad \varrho_t = \frac{\text{RV}_t}{\text{IV}_t^2},$$

  with mean $\overline{\text{ES}}_\theta$ over $\mathcal{T}_{\text{opt}}$.
- **The score $\text{ES}_{1,t}$ is positive** only when $\hat{\sigma}_t^2$ and RV_t lie on opposite sides of the implied variance IV_t^2.
- **Compounding:** the premium bought or sold each day is a fixed fraction $f = 3\%$ of current wealth, so the terminal portfolio value, from $V_0 = 1$, and the annualized log growth rate are

$$V_{T_{\text{opt}}} = \prod_{t\in\mathcal{T}_{\text{opt}}} (1 + f\,r_t), \qquad G = \frac{252}{T_{\text{opt}}}\sum_{t\in\mathcal{T}_{\text{opt}}} \log(1 + f\,r_t).$$

- At quoted bid/ask prices, r_t^{ba} replaces $r_t = q_t R_t$. Maximum drawdown $\max_t(1 - V_t/\max_{t'\le t} V_{t'})$, with V_t the portfolio value after day t.

6.3 Results

- **The unconditional short-straddle benchmark** ($q_t \equiv -1$; "Short straddle" in the tables): SR = 0.20 at mid-quotes and -0.27 at quoted bid/ask prices. Its mean return is not significantly different from zero (HAC $t = 0.43$), and it loses money in 3 of the 5 calendar years: selling the closing straddle earns no statistically detectable variance risk premium in the option-data subsample.
- **Sign strategies of the pooled intraday models versus the short-straddle benchmark** (point estimates): $\Delta\text{SR} > 0$ for all 12 pooled intraday forecasts.

  - 3 of the gains are significant at the 5% level (HAC t-statistic on the daily return differential), all three on the full predictor set: LightGBM ($t = 2.05$), XGBoost ($t = 1.99$) and the random forest ($t = 2.13$).
- **Highest Sharpe ratios** (point estimates), all on the full predictor set:
  - elastic net: SR = 1.50 at mid-quotes and 1.01 at quoted bid/ask prices (HAC $t = 1.90$ on its gain over the short-straddle benchmark);
  - ridge: 1.44 at mid-quotes, 0.96 at quoted bid/ask prices;
  - LightGBM: 1.43 at mid-quotes, 0.93 at quoted bid/ask prices.
- **Full predictor set versus the HAR benchmark:** the full predictor set raises SR at mid-quotes in all 6 model classes, from 0.69–1.01 to 1.20–1.50.
  - ΔSR from +0.30 to +0.61 (median +0.51; elastic net +0.61, ridge +0.53, lasso +0.51);
  - the gain is economically large but not significant at the 5% level in any class (HAC t-statistic on the daily return differential).
- **Position sizing,** median SR at mid-quotes across the 12 pooled intraday forecasts: sign strategy 0.90, robust clip 1.37, rank 1.40.
  - $\mathrm{SR}^{\mathrm{clip}} > \mathrm{SR}^{\mathrm{sign}}$ for 11 of the 12 forecasts (point estimates; no test).
- **The value is concentrated in a few days**:
  - without its 10 best days, the mean return differential over the short-straddle benchmark, $\bar{\Delta}$, is negative for all 12 pooled intraday forecasts;
  - days on which the package expires worthless ($R_t = -1$, settlement pins) supply most of the return.
- **Statistical accuracy is a weak guide to economic value.** Spearman rank correlations across all 112 forecasts of the model set (both training schemes), on $\mathcal{T}_{\mathrm{opt}}$, between average loss and the Sharpe ratio (negative: the more accurate forecasts earn the higher SR):

$$\rho_S(\overline{\mathrm{QLIKE}}, \mathrm{SR}) = -0.20, \qquad \rho_S(\overline{\mathrm{ES}}_1, \mathrm{SR}) = -0.55.$$

- **Robustness** (appendix):
  - shift test, $q_t = \mathrm{sign}(\hat{y}^2_{t,c+j}\, s^2_{t,c} - \mathrm{IV}^2_t)$, with the forecast of the interval j half hours away from the closing interval ($j < 0$: earlier on day t; $j > 0$: on day $t+1$): the median SR ranges from 0.00 to 1.58 across j (a swing of ± 0.79), within the ± 1.06 95% sampling band of an annualized Sharpe ratio under i.i.d. returns, so differences of this size are sampling noise;
  - subsample splits around May 2022;
  - compounding, across the 112 forecasts of the model set: median terminal portfolio value $V_{T_{\mathrm{opt}}}$ of 4.4 at mid-quotes and 1.8 at quoted bid/ask prices; maximum drawdowns of up to 64% and 78%, respectively.

6.4 A final refinement: training on the last 30 minutes

- **The scheme** (the closing-interval model, $\mathcal{I} = \{c\}$): every model class of Section 4.2 is estimated on the last 30 minutes of each trading day only, the closing interval $i = c$ (15:30–16:00 ET), as the benchmark forecast is.
  - One observation per trading day, so a period is a trading day and $\mathcal{W}_t = \{t-W, \ldots, t-1\}$ holds $W = 2{,}000$ trading days (Appendix B.2).

  - Predictor sets, evaluation span and trading rules as before; accuracy in Table 2, on the same days as Table 1.

- **Training on the last 30 minutes lowers average QLIKE further,** in every model class and on both predictor sets: the pooled intraday model's average QLIKE exceeds that of the closing-interval model of the same class, $\Delta\% = 100(\bar{L}^{\text{pool}}/\bar{L}^{\text{close}} - 1) > 0$, with $\bar{L}^{\text{pool}}$ and $\bar{L}^{\text{close}}$ the average losses of $f_{m,\mathcal{X},\{1,\ldots,M\}}$ and $f_{m,\mathcal{X},\{c\}}$:
  - linear models: +10.0 to +10.7% on the HAR benchmark (DM from 3.09 to 3.22, significant at the 5% level) and +5.3 to +11.1% on the full predictor set (significant for the lasso only);
  - trees: +18.5 to +21.7% on the HAR benchmark and +7.5 to +9.1% on the full predictor set, each significant at the 5% level.
- **The feature result holds:** the full predictor set lowers average QLIKE in all 6 closing-interval model classes, $\Delta\%$ from -4.1 to -0.1, none significantly at the 5% level (lasso: $\Delta\% = -4.0$, DM $= -1.91$, $p = 0.056$).
  - Across the 51 predictor-set pairs of the model set (45 tree and 3 linear specifications trained on the last 30 minutes, 3 pooled intraday), it lowers average QLIKE in 35, 4 of them significantly.
- **The most accurate forecasts of the paper** are closing-interval models on the full predictor set: average QLIKE lasso 0.0992, elastic net 0.1014, ridge 0.1031, XGBoost 0.1031; the benchmark forecast ($\overline{\text{QLIKE}} = 0.1033$) ranks 8 of 116.
  - Against the benchmark forecast, the lasso changes average QLIKE by -4.0% (DM $= -1.91$, $p = 0.056$); none of the four is significantly more accurate at the 5% level.
  - Of the 111 other forecasts of the model set, 7 have a lower average QLIKE than the benchmark forecast, 0 of them significantly, and 56 are significantly less accurate.
  - Trees versus the closing-interval ridge on the same predictors: the tree has the higher average QLIKE in 48 of 48 comparisons, 23 of them significantly.
- **Squared-error loss disagrees:** the full predictor set raises the MSE in 49 of the 51 pairs of the model set, 0 of them significantly.
  - 19 March 2020, a day outside the option-data subsample, carries 28–39% of the closing-interval trees' total squared error over $\mathcal{T}_{\text{oos}}$ but at most 0.9% of their total QLIKE loss.
  - Five days in March 2020 carry 73–87% of the squared error of the benchmark forecast and of the closing-interval lasso and XGBoost on the full predictor set; excluding February–April 2020, that lasso has a lower MSE than the benchmark forecast.
- **Economic value** (sign strategy on $\mathcal{T}_{\text{opt}}$; point estimates):
  - The highest Sharpe ratio of the model set: the elastic net on the full predictor set, SR = 1.92 at mid-quotes and 1.43 at quoted bid/ask prices, its gain over the short-straddle benchmark significant at the 5% level (HAC $t = 2.56$); the lasso on the full predictor set: 1.88 and 1.40. Both exceed every pooled intraday forecast (SR ≤ 1.50 at mid-quotes).
  - LightGBM, the best of its specifications by SR on $\mathcal{T}_{\text{opt}}$ (an ex post selection): 1.81.
  - Full predictor set minus HAR benchmark: ΔSR $= +0.55$ (elastic net) and +0.50 (lasso), neither significant at the 5% level; across the 51 pairs of the model set, $\Delta\text{SR} > 0$ in 50 (median +0.54), 4 of them significantly.
  - Against the short-straddle benchmark, $\Delta\text{SR} > 0$ for 112 of the 112 forecasts of the model set; 20 of the gains are significant at the 5% level, all of them with the full predictor set.

– None of the differences in SR between the pooled intraday and the closing-interval model of the same class is significant at the 5% level (HAC t-statistic on the daily return differential, $q_{\mathrm{NW}} = 14$).

Table 2: Forecast accuracy when training on the last 30 minutes

	Last 30 minutes		
Model and predictor set	MSE	QLIKE	R^2_{OOS} (%)
Panel A: Linear models			
OLS (benchmark forecast)			
HAR benchmark	4.15	0.1033	78.2
Ridge			
HAR benchmark	4.15	0.1033	78.2
All features	7.30	0.1031	61.7
Lasso			
HAR benchmark	4.15	0.1033	78.2
All features	5.94	0.0992	68.8
Elastic net			
HAR benchmark	4.15	0.1032	78.2
All features	6.89	0.1014	63.9
Panel B: Tree ensembles			
LightGBM			
HAR benchmark	5.75	0.1094	69.8
All features	17.19	0.1049	9.8
XGBoost			
HAR benchmark	5.27	0.1076	72.4
All features	20.94	0.1031	−9.9
Random forest			
HAR benchmark	4.62	0.1068	75.8
All features	14.15	0.1060	25.8

Notes: Rolling forecasts from 3 January 2020 to 30 April 2024, every forecast made with a model estimated on its trailing window only, so every forecast is out of sample; each is scored against the actual realized variance of the closing half hour, RV_t, on the $T_{\mathrm{oos}} = 1{,}082$ trading days on which every forecast of both training schemes exists, early closes dropped (Section 4.4). The span starts with the pooled intraday linear forecasts, on the first trading day after the pre-sample on which the penalties were selected (6 January 2016 to 2 January 2020). MSE is the mean squared error $T_{\mathrm{oos}}^{-1}\sum_t(\mathrm{RV}_t - \hat{\sigma}_t^2)^2$, multiplied by 10^{10}, and QLIKE the mean QLIKE loss. $R^2_{\mathrm{OOS}} = 1 - \sum_t(\mathrm{RV}_t - \hat{\sigma}_t^2)^2 / \sum_t(\mathrm{RV}_t - \overline{\mathrm{RV}}_{t-1})^2$, in percent, where the benchmark $\overline{\mathrm{RV}}_{t-1}$ is the historical mean of the closing-half-hour realized variance over all trading days from 5 January 1998 to day $t-1$ (Campbell and Thompson, 2008). All features is the full predictor set of Section 3. Each model is estimated on the last 30 minutes of each trading day only (one observation per day); the OLS on the HAR benchmark is the benchmark forecast. Trees use fixed hyperparameters and are re-estimated daily (Section B.2). The pooled intraday models on the same days are in Table 1.

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A Appendix structure

- **Implementation details** (Appendix B): samples and windows, the training schemes in practice, the tuned hyperparameters and the tuning procedures.
- **A. Data and predictors:**
  - sources and spans;
  - predictor dictionary: 47 exogenous series $\rightarrow$ 628 predictors in the full predictor set;
  - missing and stale data;
  - removal of duplicate predictors; the window filter (constant and duplicate predictors dropped in each estimation window $\mathcal{W}_t$);
  - SPXW option chain; implied-volatility inversion;
  - FOMC calendar.
- **B. Hyperparameter selection by model:**
  - linear models: penalty strength λ, elastic-net mixing α; block-penalized ridge, penalty weights ω_j;
  - tree ensembles: fixed hyperparameters, Optuna budgets, selected trials; boosting rounds $b = 1, \ldots, B$, learning rate η;
  - re-selection frequency, on the estimation sample $\mathcal{W}_t^{\text{est}}$, gap $\mathcal{W}_t^{\text{gap}}$ and validation sample $\mathcal{W}_t^{\text{val}}$; re-estimation frequency;
- **C. Robustness, statistical:**
  - full model table, tree ensembles included;
  - Diebold–Mariano and HAC conventions (DM with the Harvey–Leybourne–Newbold correction; Newey–West lag truncation q_{NW});
  - model confidence set, superior predictive ability, StepM;
  - regime and break tests;
  - closing-interval vs pooled intraday models;
  - remaining variance through the trading day.
- **D. Robustness, trading:**
  - date-shift test;
  - Murphy diagrams (elementary scores); subsample splits;
  - P&L decomposition;
  - compounding;

  - position sizing;
  - strategy variations; other forecast origins; month-ends.
- **E. Reproducibility:**
  - software lock files;
  - hashes of the design matrices;
  - computing environment;
  - CPU classes;
  - compute time;
  - the generator script behind every reported number.

B Implementation details

B.1 Samples and windows

- **Estimation and evaluation samples:** Table 3.
- **Burn-in:** the first 3,125 intervals, the longest HAR horizon, are dropped (Section 2).
- **Diurnal adjustment** $s^2_{t,i}$ (Section 4.1): $n_{\mathrm{d}} = 20$ trading days, at least $\underline{n}_{\mathrm{d}} = 5$ of them; intervals with fewer are not scored.
- **Winsorization** of the target: $n_{\mathrm{w}} = 240$ intervals, all intervals of the day pooled; the bounds are the empirical 1st and 99th percentiles, $\underline{Q}_{t,i} = Q_{0.01}(\mathcal{Y}_{t,i})$ and $\overline{Q}_{t,i} = Q_{0.99}(\mathcal{Y}_{t,i})$.
- **Hyperparameter-selection splits:** $n_{\mathrm{gap}} = 25$ and $n_{\mathrm{val}} = 125$ observations.

B.2 Training schemes in practice

- **Closing-interval scheme:** $W = 2{,}000$ trading days for every model class; re-estimated every trading day, except the tree specifications re-estimated every 10 trading days (fixed hyperparameters or random search).
- **Pooled intraday scheme:** $M = 48$ intervals a day; the full predictor set has 1,144 predictors (Section 3).
  - Linear models: $W = 96{,}000$ intervals (2,000 trading days), re-estimated every interval.
  - Trees: $W = 24{,}000$ intervals; one fixed hyperparameter configuration per tree method, never re-selected, re-estimated every interval.

B.3 Tuned hyperparameters

- **Linear penalties,** chosen by hyperparameter selection on a pre-sample (Section B.4), one grid value per decade (Table 4):

$$\lambda^{\mathrm{ridge}} \in \{10^{-2}, \ldots, 10^{3}\}, \qquad \lambda^{\mathrm{lasso}}, \lambda^{\mathrm{enet}} \in \{10^{-6}, \ldots, 10^{-2}\}.$$

  - full predictor set: ridge 10^3, lasso and elastic net 10^{-2} (top edge);
  - HAR benchmark: the lower edge of each grid.
- **Trees,** fixed hyperparameters from a search on a 1998–2002 pre-sample (Table 4):
  - LightGBM and XGBoost: the best trial of an Optuna search (Akiba et al., 2019);
  - random forest: trial 255 of 400 ($B = 485$ trees, maximum depth 20, share of predictors tried at each split 0.90, bootstrap sample share 0.41);

– leaf minimums are rescaled to the 2,000-observation window: $n_{\min} \leftarrow n_{\min} \cdot 2{,}000/24{,}000$, the pooled intraday trees' window (Section B.2).

B.4 Tuning procedures

- **Linear:**
  - hyperparameter selection on a pre-sample of four folds of 250 trading days (6 January 2016 to 2 January 2020), on validation MSE:

$$\hat{\lambda} = \arg\min_{\lambda} \frac{1}{4} \sum_{v=1}^{4} \mathrm{MSE}_v^{\mathrm{val}}(\lambda),$$

    with $\mathrm{MSE}_v^{\mathrm{val}}(\lambda)$ the mean squared error of $\tilde{y}$ on the validation sample of fold v;
  - a light re-tune at each re-tuning date (every 250 trading days): the grid neighbors $\mathcal{C} \subseteq \{\lambda^-, \lambda^+\}$ (one decade below and above, where on the grid) challenge the incumbent λ;
  - the best challenger λ' (lowest validation MSE) replaces it only if the Diebold–Mariano statistic DM of Section 4.5, on the paired squared errors, is significantly negative:

$$\mathrm{DM}(\{e_{t'}^2(\lambda') - e_{t'}^2(\lambda)\}_{t'}) < -z_{0.05/|\mathcal{C}|}, \qquad z_{0.05/|\mathcal{C}|} = \Phi^{-1}(1 - 0.05/|\mathcal{C}|);$$

  - $e_{t'}(\lambda) = \tilde{y}_{t'} - \hat{y}_{t'}(\lambda)$, t' over two 125-day validation folds; Φ: the standard normal distribution function.
- **Trees,** in increasing order of tuning effort:
  - fixed hyperparameters, re-estimated every 10 trading days or daily;
  - random search every 250 trading days;
  - Optuna (Akiba et al., 2019): 50 trials, re-tuned every 1, 5, 25 or 250 trading days, best of the first 10, 25 or 50 trials.
- **Every selection uses only data available at the forecast origin;** no forecast is scored on a day that entered the selection of its own hyperparameters, and the evaluation span $\mathcal{T}_{\mathrm{oos}}$ starts after the pre-sample.

C Notes to the authors (not part of the paper)

- **The current title, abstract, introduction and conclusion describe the old 24-hour panel** and need replacing.
- **The results text anchors on the pooled block ridge;** re-base it on the per-bar OLS. Against that OLS, 7 of 111 forecasts have lower QLIKE, none significantly.
- **Data issue to fix before final numbers:**
  - the activity flags are computed after the forward fill, so the FOMC flags read 1 on all 122 trade days after November 2023;
  - VIX, volume-demand and FOMC feeds end early and are carried forward without limit into 2024.
- **Missing outputs:** paired-comparison tables (axis 1, axis 2), DM against the OLS in the master table, and the model confidence set over models.
- **Pending:** the pooled random forest (full table).

Table 3: Estimation and evaluation samples.

Sample	Start	End	Obs.	Window W
Intraday grid: 30-minute intervals with an observed $\mathrm{RV}_{t,i}$, outside the weekend break	1998-01-05	2024-04-30	303,442	–
Panel A: Closing-interval models (trading days)				
Closing-interval observations, after the burn-in	2010-07-12	2024-04-30	3,469	2,000
Rolling-window forecasts	2018-06-25	2024-04-30	1,469	–
Hyperparameter-selection pre-sample (linear models)	2004-04-13	2020-01-02	3,947	2,000
Validation samples (4×250 days)	2016-01-06	2020-01-02	1,000	2,000
Panel B: Pooled intraday models (30-minute intervals)				
Estimation sample, after the 3,125-interval burn-in	1998-06-04	2024-04-30	300,317	24,000
Rolling-window forecasts	2001-07-13	2024-04-30	276,317	–
Tree hyperparameter pre-sample (before 2003)	1998-06-04	2002-12-31	36,285	24,000
Panel C: Evaluation samples (trading days)				
Evaluation span (T_{oos}): trading days on which every forecast exists, less early-close days (6)	2020-01-03	2024-04-30	1,082	–
Option-data subsample (T_{opt}): SPXW expiration days (872) less early-close days (6)	2020-01-03	2024-04-30	866	–
Later SPXW expiration days, held out (413 full trading days)	2024-05-01	2025-12-31	419	–

Notes: Dates are interval-end dates (ET); Obs. and the rolling estimation window W are in the units of each panel. The burn-in, 3,125 intervals, is the longest HAR horizon and is applied to each series as built (the closing-interval series, the intraday grid and, for the linear pre-sample, the series of the two intervals of the last hour), so the closing-interval sample starts in 2010 and the pre-sample in 2004. The pooled linear models use $W = 96{,}000$ intervals (2,000 trading days); the tree pre-sample is re-estimated every 250 intervals. Each validation sample follows a 25-day gap and a 2,000-day estimation sample (Section B.1). Full trading days: expiration days that are not early-close days. Panel C: the evaluation span holds every trading day on which all rolling forecasts exist (Section 4.4); the straddle economics use its option-data subsample (Section 6).

Table 4: Hyperparameters: penalized regressions and tree ensembles.

			Value	
Model	Hyperparameter	Search space	HAR benchmark	Full predictor set
Panel A: Penalized linear models, selected on the pre-sample				
OLS	–	–	–	–
Ridge	Penalty, λ	$\{10^{-2}, 10^{-1}, \ldots, 10^{3}\}$ (6)	10^{-2}	10^{3}
Lasso	Penalty, λ	$\{10^{-6}, 10^{-5}, \ldots, 10^{-2}\}$ (5)	10^{-6}	10^{-2}
Elastic net	Penalty, λ	$\{10^{-6}, 10^{-5}, \ldots, 10^{-2}\}$ (5)	10^{-6}	10^{-2}
	Mixing weight, α	fixed	0.5	0.5
All three	Penalty weight on the HAR benchmark, $\underline{\omega}$	fixed	1	0.01
Panel B: Tree ensembles, fixed hyperparameters (selected on the 1998–2002 pre-sample)				
LightGBM	Leaves per tree, J	[15, 255] (log)	20	
	Learning rate, η	[0.01, 0.3] (log)	0.013	
	Boosting rounds, B	[50, 500]	392	
	Minimum leaf size, $n_{\min}$	[20, 500] (log)	98 (8)	
	Share of predictors per tree	[0.5, 1]	0.70	
	Share of observations per tree	[0.5, 1]	0.56	
	L_1 penalty on leaf weights, λ_1^{tree}	$[10^{-8}, 10]$ (log)	10^{-8}	
	L_2 penalty on leaf weights, λ_2^{tree}	$[10^{-8}, 10]$ (log)	0.016	
XGBoost	Maximum depth	[3, 10]	5	
	Learning rate, η	[0.01, 0.3] (log)	0.0123	
	Boosting rounds, B	[50, 500]	367	
	Minimum leaf size, $n_{\min}$	[1, 100] (log)	50.2 (4.18)	
	Share of observations per tree	[0.5, 1]	0.66	
	Share of predictors per tree	[0.5, 1]	0.65	
	L_1 penalty on leaf weights, λ_1^{tree}	$[10^{-8}, 10]$ (log)	0.0215	
	L_2 penalty on leaf weights, λ_2^{tree}	$[10^{-8}, 10]$ (log)	1.2×10^{-5}	
	Penalty per leaf, γ	$[10^{-8}, 1]$ (log)	1.5×10^{-6}	
Random forest	Number of trees, N	[50, 500]	485	
	Maximum depth	[2, 32] (log)	20	
	Share of predictors per split	[0.02, 1] (log)	0.90	
	Bootstrap sample share	[0.05, 1] (log)	0.41	
	Minimum leaf size, $n_{\min}$	[1, 1000] (log)	12 (1)	

Notes: Panel A: λ is on the scale of the objectives in Section 4.2, one grid value per decade (grid size in parentheses), selected by mean validation MSE over the 4 folds of the pre-sample (Table 3). At each of the 4 re-selection points a grid neighbor may replace it (Section B.4); this happened 4 times in 24 re-selections (ridge 1 and 0, lasso 1 and 0, elastic net 2 and 0, for the HAR benchmark and the full predictor set). Panel B: the best trial of an Optuna search on the pooled intraday pre-sample (24,000-interval window; random forest: trial 255 of 400), ranked by MSE and then held fixed for both predictor sets; (log): searched on a logarithmic scale. $n_{\min}$ was selected on 24,000 intervals and is rescaled to the 2,000-day window (rescaled value in parentheses). Implementations: LightGBM (Ke et al., 2017) with bagging at every round, XGBoost (Chen and Guestrin, 2016) with histogram-based split finding and the scikit-learn random forest with bootstrap samples; random seeds are fixed.

- **The full outline,** with source and macro pointers, is `writeup/PAPER_REHAUL_OUTLINE_2026-10-15.md`.